

Design and Development of an Affordable Refreshable Braille Display Using an Embedded Microcontroller

ABSTRACT

This project presents the design and development of an affordable refreshable Braille display module using an embedded microcontroller. The system provides an accessible digital interface for visually impaired individuals by dynamically generating tactile Braille characters, effectively bridging the gap in digital accessibility. Designed to overcome the prohibitive cost of traditional piezoelectric-bimorph displays, the module utilizes an alternative, low-cost electromechanical actuation mechanism. When digital text payloads are received via a standard communication interface, the embedded microcontroller manages precise timing, spatial coordination, and power distribution to accurately elevate and retract specific Braille pins in real-time. The proposed design offers a highly scalable, low-cost, and reliable solution for rendering dynamic tactile information in educational institutions, libraries, and personal environments, eliminating the reliance on expensive proprietary hardware. Future enhancements may include expanded multi-cell display architectures, wireless Bluetooth connectivity for direct smartphone integration, and optimized battery-powered portability for improved independent reading and continuous communication.

Design and Development of an Intelligent Bearing Fault Early-Warning System Using an Embedded Microcontroller

ABSTRACT

This project presents the design and development of an intelligent predictive maintenance system for rotating machinery using an embedded microcontroller with high-frequency vibration sensing. The system is designed to protect valuable industrial assets by identifying early-stage bearing degradation and mechanical imbalance before unexpected catastrophic failure occurs. Powered efficiently for continuous edge deployment, a motion and vibration sensor continuously monitors the protected equipment's mechanical signatures. When anomalous vibration patterns are detected through local spectral analysis, the microcontroller immediately triggers an audible alarm and real-time diagnostic alerts, actively filtering out ambient background noise from adjacent factory equipment. The proposed design offers an effective, affordable, and spatially isolated approach to machine health monitoring suitable for small and medium-sized manufacturing enterprises, workshops, and automated assembly lines. Future enhancements may include advanced multi-sensor data fusion, wireless notification capabilities through industrial communication protocols, and automated fleet-level health dashboards for remote predictive analytics.

Design and Development of a Self-Correcting Smart Soil Monitoring System Using an Embedded Microcontroller

ABSTRACT

This project presents the design and development of an intelligent and self-correcting soil monitoring system using an embedded microcontroller integrated with multi-parameter environmental sensors. The system addresses the fundamental problem of long-term measurement drift in affordable capacitive soil moisture sensors caused by fluctuating soil salinity and temperature extremes. Powered efficiently for remote agricultural deployment, the module continuously captures co-located electrical conductivity, temperature, and baseline moisture data. When shifting environmental conditions are detected, the microcontroller utilizes dynamic, regression-based algorithms to compensate for sensor interference in real-time, ensuring highly accurate moisture estimation without the need for frequent manual recalibration. The proposed design offers a low-cost, offline, and highly reliable solution for optimizing precision irrigation schedules in smallholder farms, agricultural cooperatives, and research greenhouses. Future enhancements may include autonomous water valve control, wireless mesh networking for expansive field coverage, and AI-assisted predictive irrigation forecasting based on aggregated long-term weather patterns.

Design and Development of a Secure Boot and Firmware Verification System Using an Embedded Microcontroller

ABSTRACT

This project presents the design and development of a hardware-anchored secure boot and firmware verification system using a resource-constrained embedded microcontroller. The system provides a critical layer of cybersecurity by continuously authenticating firmware payloads before permitting application execution on vulnerable embedded and IoT devices. Operating directly at the hardware-firmware boundary, the system ensures that only digitally signed and explicitly authorized code can successfully boot. When the device is powered on or receives an over-the-air (OTA) update, the embedded microcontroller validates the incoming cryptographic signature against trusted keys, instantly rejecting unauthorized, spoofed, or tampered firmware payloads while triggering a local security alert. The proposed design offers a low-cost, verifiably correct security architecture suitable for hardening industrial controllers, medical devices, and consumer IoT networks against supply-chain compromises. Future enhancements may include post-quantum cryptographic verification algorithms, remote attestation reporting to centralized servers, and automated integration with enterprise device-management platforms.

Design and Development of a Multi-Modal Cold-Chain Integrity Logger Using an Embedded Microcontroller

ABSTRACT

This project presents the design and development of a multi-modal cold-chain integrity logger using a low-power embedded microcontroller equipped with thermal and inertial sensors. The system is designed to enhance the logistics of temperature-sensitive pharmaceuticals, vaccines, and perishables by continuously monitoring the genuine physical state of the cargo. Powered entirely by an energy-efficient internal source for long-haul tracking, the module calculates the actual core temperature of the liquid payload using sophisticated thermal-mass modeling. When a sustained thermal excursion or a sudden mechanical shock is detected, the microcontroller logs a compound damage event, intelligently ignoring transient air-temperature spikes caused by routine door openings. The proposed design offers a highly reliable, low-cost solution for optimizing supply chain accountability in medical distributions and complex food logistics without relying on continuous cloud connectivity. Future enhancements may include real-time cellular alerting mechanisms, GPS location stamping, and predictive spoilage modeling based on accumulated environmental stress.